

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	08 July 2026
Team ID	xxxx
Project Name	Cosmetics Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	- Registration through Form - Registration through Gmail - Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Data Ingestion & Aggregation Pipeline	Automated daily ingestion of multi-channel cosmetics industry data (social media sentiment, e-commerce reviews, search volume trends) Data transformation, cleaning, and preparation layer to structure unstructured consumer sentiment metrics
FR-4	Interactive Tableau Dashboard & Insight Analytics	Dynamic trend visualization matrices capturing shifting ingredient preferences, brand affinity, and dynamic geometric scaling metrics Advanced demographic and category filters (age bracket, geographic region, price tiers, and skincare vs. makeup segmenting)

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The Tableau dashboard interface must be intuitive, requiring zero formal training for brand managers, utilizing clear interactive legends and tooltips.
NFR-2	Security	Role-based access controls (RBAC) enforced via Tableau Server authentication to protect sensitive, proprietary enterprise market analytics.
NFR-3	Reliability	Data synchronization pipeline must guarantee 99.9% fault-tolerant execution, with built-in rollback mechanisms to prevent dashboard data corruption.
NFR-4	Performance	Dashboard response time for complex filter selections and view switches must render in under 3 seconds using pre-computed hyper-extracts
NFR-5	Availability	The hosted analytics dashboard and pipeline monitoring environment must maintain 24/7 global availability with minimal scheduled off-peak maintenance.
NFR-6	Scalability	The data storage and visualization tier must scale seamlessly to support a 200% increase in data row volume and up to 100 concurrent dashboard viewers without latency degradation.